Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software developer with 2+ years building **C++/Python applications, concurrent systems, and financial platforms**; strong in Linux, data structures, algorithms, mathematical modeling, performance optimization, distributed systems, user-facing interfaces, and collaborative problem-solving.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Engineered **concurrent distributed Python services** processing 1,000+ jobs with explicit state transitions, failure isolation, and optimized data access, increasing throughput 3x while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Designed PostgreSQL and SQL Server transformations for 10,000+ records, applying indexing, validation, and query optimization to reduce latency 35% while sustaining 99%+ reliability across recurring production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built a **multithreaded Python processing engine** using ThreadPoolExecutor, secure file ingestion, and fault-tolerant execution, handling 1,000+ files per batch and 10,000+ daily requests with stable peak-load performance.
- Collaborated closely with engineers through design discussions, code reviews, testing, and CI/CD releases, integrating multiple model providers behind adaptable interfaces while reducing user feedback turnaround time 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Translated **mathematical models into efficient Python software** for autonomous-driving research, building large-scale data and simulation pipelines over 100,000+ trajectory logs and improving long-horizon planning performance 18%.
- Designed a spatial-temporal model combining road geometry, lane direction, and historical trajectories through multi-head attention, improving closed-loop stability 15% while evaluating quantitative metrics across 10,000 driving episodes.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Optimized **cross-platform production software and algorithms** across Linux, Windows, and macOS using OOP patterns, SQL, caching, and API integrations, reducing response times 30% and technical debt 20%.
- Built reliable development and test automation with Git, Jenkins, PyTest, and Mockito, executing 20+ tests and parallel builds while collaborating with Microsoft and GitHub engineers across Agile sprints.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Developed **secure financial application services and interfaces** for pricing, payments, refunds, transaction status, and analytics, supporting 50,000+ records with real-time Firebase synchronization and sub-200-millisecond API responses.
- Implemented secure multi-currency transaction workflows across cards, wallets, and webhooks, automating refund and dispute handling by 70% while documenting 12+ interfaces and presenting system designs to clients.

Skills

Languages: C++, Python, Java, SQL, Bash, JavaScript, TypeScript
Computer Science: Data structures, algorithms, OOP, design patterns, multithreading, concurrency, memory, I/O
Systems: Linux, distributed systems, real-time pipelines, performance optimization, caching, REST APIs, WebSockets
Data & Finance: PostgreSQL, SQL Server, MongoDB, Firebase, transaction systems, payments, pricing, analytics
Engineering: Git, Jenkins, GitHub Actions, CI/CD, Docker, PyTest, testing, debugging, code review, Agile collaboration

Projects

CPU Scheduling Simulator Live | *C++, Algorithms, Systems*

- Implemented and benchmarked **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.

FinNexus Live | *Financial Systems, TypeScript, MongoDB*

- Built a **real-time financial platform** supporting 50,000+ users and 1M+ transactions with secure bank integrations, token-based access, advanced filtering, MongoDB persistence, and Kubernetes-based distributed deployment automation.

Chat2SQL Code | *Java, TypeScript, Linux, Docker*

- Enhanced a **cross-platform database application** across Java and TypeScript components, supporting 10+ SQL and NoSQL systems with schema exploration, query generation, Docker, Maven, Linux workflows, and reusable interfaces.